

1. Administrative & Study Identification

Full Study Title: A Phase I/II, Open-label, Dose Escalation and Dose Expansion Study to Evaluate Safety, Pharmacokinetics, Pharmacodynamics and Efficacy of AZD2936 Anti-TIGIT/Anti-PD-1 Bispecific Antibody in Participants With Advanced or Metastatic NSCLC **Short Title/Acronym:** ARTEMIDE-01 **Protocol Number:** D9180C00001 (Internal Study ID) **NCT Number:** NCT04995523 **Sponsor Name:** AstraZeneca **Investigational Product:** Rilvegostomig (AZD2936) – Anti-TIGIT/Anti-PD-1 Bispecific Antibody **Phase of Development:** Phase I/II **Medical Monitor:** AstraZeneca Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate if the experimental anti-TIGIT/anti-PD-1 bispecific antibody rilvegostomig (AZD2936) is safe, tolerable, and efficacious, assessing dose-limiting toxicities (DLTs) and safety profiles in participants with Advanced or Metastatic Non-small Cell Lung Cancer (NSCLC). **Secondary Objectives:** To evaluate the Objective Response Rate (ORR), Disease Control Rate (DCR), Duration of Response (DoR), pharmacokinetics (PK), and pharmacodynamics (PD), defined as percentage PD-1 and TIGIT receptor occupancy (RO).

2.2 Background & Rationale PD-1 and TIGIT are heavily implicated in cancer-related T cell immunosuppression. Rilvegostomig (AZD2936) is designed to provide a dual blockade approach that seeks to potentiate T-cell activation and overcome immune escape mechanisms. This trial aims to address the clinical necessity for novel therapies that can produce more durable responses in advanced solid tumors for patients who have progressed on checkpoint inhibitors (CPI-experienced) or are CPI-naïve.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm (Multi-part: Dose-escalation [Part A] and Dose-expansion [Parts B-E]) **Blinding:** Open-label **Randomization:** N/A (Sequential cohort assignment)

3.2 Planned Enrollment & Duration Target \$N\$: Approximately 80-120 evaluable patients required across the dose-escalation and expansion cohorts. **Number of Sites:** Multicenter (Global) **Estimated Study Start:** August 2021 **Estimated Primary Completion:**

Approximately 2 years per participant follow-up (Part A-E ongoing timeline).

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Aged 18 or above. 2. Documented PD-L1 expression by PD-L1 IHC per local report. 3. Unresectable stage III or stage IV squamous or non-squamous NSCLC not amenable to curative surgery or radiation (Parts A-D) or Stage IV squamous NSCLC (Part E). Part A/B must have confirmed progression during treatment with a CPI-including regimen; Part C/D/E must be CPI-naïve.

4.2 Exclusion Criteria 1. Previous treatment with an anti-TIGIT therapy. 2. Any concurrent chemotherapy, radiotherapy, investigational, biologic, or hormonal therapy for cancer treatment. 3. Symptomatic central nervous system (CNS) metastasis or a thromboembolic event within 3 months prior to enrollment.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Rilvegostomig administered Intravenously every 3 weeks (Q3W). Dose escalation ranged from 70 mg to 1500 mg, establishing a Recommended Phase 2 Dose (RP2D) of 750 mg for expansion cohorts. **Route of Administration:** Intravenous (IV) Infusion. **Duration of Treatment:** Treatment cycles continue until disease progression, unacceptable toxicity, or death in the absence of disease progression (approx. 2 years).

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for advanced oncology patients. **Prohibited:** Concurrent administration of chemotherapy, investigational agents, immunotherapy (other than the study drug), or biological cancer treatments.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Baseline Scans, PD-L1 IHC testing
Baseline	Day 0	First IV Dose (Cycle 1, Day 1), PK/PD blood draws
Interim	Every 3 Weeks (Q3W)	Safety Labs, Efficacy Assessment (RECIST tumor imaging), IV infusion
End of Study	Variable	Final Vital Signs, Exit Interview, Safety follow-up post-disease progression

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Incidence of Dose-Limiting Toxicities (DLTs) and Treatment-Related Adverse Events (TRAEs) from the first dose up to the end of the DLT evaluation period. **Measurement Method:** Clinical safety evaluation, laboratory tests (e.g., lipase evaluation), and WHO Toxicity grading.

7.2 Secondary & Exploratory Endpoints 1. Objective Response Rate (ORR), Disease Control Rate (DCR), and Duration of Response (DoR). 2. Pharmacokinetics (PK) and Pharmacodynamics (PD), specifically evaluating the percentage of PD-1 and TIGIT receptor occupancy.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Patient monitoring prior to each Q3W dosing cycle. Specific tracking for known immune-related AEs (e.g., pruritus, rash, and increased lipase). **Serious Adverse Events (SAEs):** Required reporting within 24 hours of investigator awareness. **Data Monitoring Committee (DMC):** Internal Safety Review Committee evaluating dose escalation progression and establishing the RP2D.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set for safety and DLT evaluations; Intent-to-Treat/Efficacy evaluable population for tumor response evaluations. **Sample Size Justification:** Adaptive trial design scaled appropriately to identify DLT thresholds and evaluate preliminary efficacy rates in specific NSCLC/PD-L1 TPS subgroups. **Handling of Missing Data:** Standard oncology trial protocols utilizing right-censoring for time-to-event endpoints (e.g., PFS, DoR) and Last Observation Carried Forward (LOCF) where applicable.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Scheduled onsite monitoring and remote data verification. **Audit Trail:** Robust centralized eCRF management and data clarification protocols maintained by the Sponsor.

11. Study Identifiers & Governance

Primary ID: ARTEMIDE-01 **Registry Number:** NCT04995523 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** ARTEMIDE-01 **Official**

Regulatory Title: A Study of AZD2936 Anti-TIGIT/Anti-PD-1 Bispecific Antibody in Participants With Advanced or Metastatic NSCLC

12. Clinical Framework (NSCLC Specific)

Primary Condition: Advanced or Metastatic Non-Small Cell Lung Cancer (NSCLC) **Diagnosis Threshold:** Stage III unresectable or Stage IV, PD-L1 expression $\geq 1\%$ **Medical Methodology:** Dual Immune Checkpoint Blockade Therapy **Optimization Target:** PD-1 and TIGIT Receptor Occupancy / Durable Tumor Response

13. Study Procedures & Methodology

Management Pathway: First-in-human oncology investigational pathway (dose escalation to dose expansion) **Education Protocol (HCP):** Investigator meeting, protocol training, and AE management training **Education Protocol (Patient):** Onsite training, informed consent, and infusion schedule coordination **Quality Audit Mechanism:** Centralized tumor scan readings and RECIST 1.1 progression tracking

14. Observation & Follow-Up Schedule

Total Duration: From initial dosing to disease progression or death (Approximately 2 years) **Onsite Visit Cadence:** Every 3 Weeks (Q3W) mirroring the IV administration schedule **Remote Monitoring Frequency:** Follow-ups conducted between clinical visit windows **Checklist Review Interval:** Regular tumor assessments every 6 to 9 weeks (Standard Oncology RECIST Interval)

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				